# PROBLEM STATEMENT

## Advertising Sales Prediction Using Multiple Linear Regression

The objective of this project is to develop a statistical and predictive model that can estimate product Sales based on advertising expenditure across three channels: **TV, Radio, and Newspaper**.

The dataset contains historical advertising spending and the corresponding Sales values. The problem is to determine how strongly each advertising channel is associated with Sales, quantify the individual contribution of TV, Radio, and Newspaper while controlling for the other channels, and build a model that can predict Sales for new advertising-budget combinations.

A reliable solution should not only fit a regression equation, but should also evaluate its predictive performance on unseen data, examine statistical significance, compare the relative importance of the predictors, and check whether the model assumptions and residual behavior are reasonable.

**Core problem: Given TV, Radio, and Newspaper advertising investments, predict Sales accurately and identify which advertising factors provide the strongest useful information for decision-making.**

## Project Questions

1. How are TV, Radio, and Newspaper advertising expenditures related to Sales?
2. What regression equation best predicts Sales from these advertising variables?
3. How accurately does the model perform on unseen test data?
4. Which predictors have statistically significant effects after controlling for the others?
5. Can an interaction between TV and Radio improve the predictive performance of the baseline model?

## Expected Outcome

The final analysis should provide an interpretable regression model, quantitative evaluation metrics, diagnostic evidence, a practical sales prediction for a proposed advertising budget, and an evidence-based comparison between the baseline model and an enhanced interaction model.

# ADVERTISING SALES PREDICTION

Enhanced Multiple Linear Regression Analysis

A detailed, notebook-grounded report that explains every executed stage of the submitted Jupyter Notebook—from data loading and exploratory analysis to regression modeling, statistical significance, diagnostics, prediction, and an empirically tested interaction model.

| | | | | |
|---|---|---|---|---|
| 200 | 4 | 80 / 20 | 0.8994 | 97.43% |
| Rows | Columns | Train / Test | Baseline Test $R^2$ | Interaction $R^2$ |

**Important source-consistency note: the existing five-page PDF does not match the current executed notebook on the regression equation and prediction. This report uses the notebook's actual executed outputs as the primary source, rather than copying the older PDF figures.**

Source files reviewed: *multiple linear regression.ipynb* and *Advertising_Sales_Report (1).pdf*.

# 1. Executive Summary

The notebook builds a Multiple Linear Regression model to predict **Sales** from three advertising investments: **TV, Radio, and Newspaper**. It uses 200 observations, keeps 20% (40 observations) completely aside for testing, and trains the baseline model on 160 observations.

The executed baseline model is **Sales = 2.9791 + 0.0447×TV + 0.1892×Radio + 0.0028×Newspaper**. On unseen test data, the notebook reports **$R^2$ = 0.8994**, meaning roughly **89.94%** of the variation in test-set sales is explained by the three predictors together.

For the planned advertising budget of TV = 150, Radio = 20 and Newspaper = 30, the notebook calculates predicted sales of **13.56 thousand units**, or approximately **13,555 units**.

A major strength of the notebook is that it goes beyond a basic regression: it compares raw and standardized coefficients, tests statistical significance with OLS, visualizes residuals, and experimentally adds a TV×Radio interaction term. The interaction model reaches **97.43% $R^2$** with RMSE 0.9015, versus 89.94% $R^2$ and RMSE 1.7816 for the baseline.

**Overall assessment: the notebook is substantially more analytically complete than the older PDF, because it includes statistical inference and a verified model improvement rather than only a baseline model.**

# 2. Notebook Workflow — Cell-by-Cell Map

| Cell | Notebook stage | What it accomplishes |
|---|---|---|
| 1 | Imports | Loads NumPy, pandas, Matplotlib, Seaborn, scikit-learn and statsmodels; configures plotting. |
| 2 | Data loading | Loads Advertising.csv with local/online fallback and standardizes column names. |
| 3 | Data quality | Checks schema, missing values and descriptive statistics. |
| 4 | Correlation | Measures pairwise relationships and visualizes the correlation matrix. |
| 5 | Train/test split | Creates X/y and reserves 20% for unseen evaluation using random_state=42. |
| 6 | Model training | Fits LinearRegression and extracts the intercept and three coefficients. |
| 7 | Test predictions | Predicts unseen test observations and calculates individual prediction errors. |
| 8 | Scenario prediction | Predicts sales for the planned advertising budget and shows the arithmetic. |
| 9 | Metrics | Calculates MAE, MSE, RMSE, test/train $R^2$ and adjusted $R^2$. |
| 10 | Statistical analysis | Adds standardized betas and OLS p-values/t-statistics. |
| 11 | Diagnostics | Plots actual-vs-predicted values and residuals. |
| 12 | Relationships | Plots each advertising channel against Sales and compares coefficients. |
| 13 | Model improvement | Adds TV×Radio interaction and verifies whether predictive performance improves. |

# 3. Data Loading, Cleaning and Structure

The data-loading cell is deliberately defensive. It first checks for **advertising.csv**, then **Advertising.csv**, and only if neither exists does it fetch the dataset from the specified online repository. It then strips whitespace, removes columns whose names contain “unnamed”, and standardizes the four expected column names to TV, Radio, Newspaper and Sales.

The executed notebook contains **200 rows and 4 columns**. All four variables are stored as floating-point values. The missing-value check returns zero missing values for every column, so the notebook does not require

imputation or row deletion.

| Variable | Role | Mean | Std. dev. | Min | Median | Max |
|---|---|---|---|---|---|---|
| TV | Predictor | 147.04 | 85.85 | 0.7 | 149.75 | 296.4 |
| Radio | Predictor | 23.26 | 14.85 | 0.0 | 22.90 | 49.6 |
| Newspaper | Predictor | 30.55 | 21.78 | 0.3 | 25.75 | 114.0 |
| Sales | Target | 14.02 | 5.22 | 1.6 | 12.90 | 27.0 |

**Interpretation:** TV has the widest spending range and the largest standard deviation. Radio has a narrower range, while Newspaper contains some comparatively high values (maximum 114). Sales ranges from 1.6 to 27.0 thousand units.

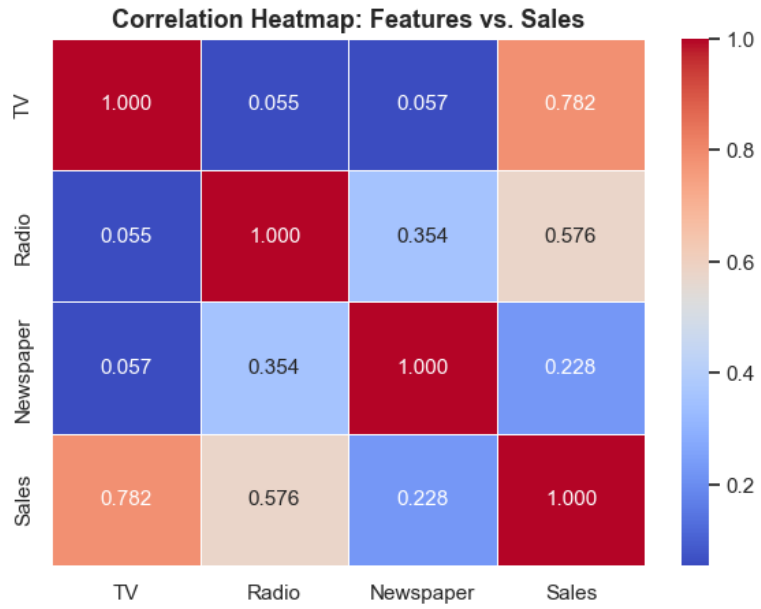
**Why this matters:** before modeling, checking data types and null values reduces the risk that a regression fails or silently learns from malformed inputs. Descriptive statistics also provide a baseline for interpreting coefficient sizes and spotting unusually large ranges.

# 4. Exploratory Correlation Analysis

The notebook computes Pearson correlations for all variables and sorts the correlation with Sales. The results are: TV **0.7822**, Radio **0.5762**, and Newspaper **0.2283**.

| Feature | Correlation with Sales | Practical reading |
|---|---|---|
| TV | 0.7822 | Strong positive linear association in this dataset. |
| Radio | 0.5762 | Moderate positive association. |
| Newspaper | 0.2283 | Weak positive association. |

The heatmap reinforces the ranking visually. Importantly, correlation is not the same as a regression coefficient: correlation is a pairwise association, whereas the regression coefficients measure the expected change in Sales for one predictor while holding the other predictors constant.




Notebook output: correlation heatmap generated in the exploratory-analysis cell.

# 5. Train/Test Design and Baseline Regression

The notebook defines X = [TV, Radio, Newspaper] and y = Sales. It then calls train_test_split with **test_size=0.20** and **random_state=42**. This produces 160 training observations and 40 unseen test observations.

The split is essential because evaluating the model on the same data used to fit it would give an overly optimistic view of predictive performance. The test set acts as a simple simulation of future observations.

The baseline model is scikit-learn's LinearRegression. Its fitted equation is:

**Sales = 2.9791 + 0.0447×TV + 0.1892×Radio + 0.0028×Newspaper**

| Term | Coefficient | Interpretation |
|---|---|---|
| Intercept | 2.9791 | Predicted Sales when all three advertising inputs are zero; mainly a mathematical baseline. |
| TV | 0.0447 | Holding Radio and Newspaper constant, one unit of TV spending is associated with +0.0447 thousand Sales units. |
| Radio | 0.1892 | Holding the other predictors constant, one unit of Radio spending is associated with +0.1892 thousand Sales units. |
| Newspaper | 0.0028 | Very small raw marginal contribution after controlling for TV and Radio. |

**Important:** coefficient magnitude alone should not be used to declare a variable “most important” when predictors have different units/ranges. The notebook correctly addresses this later with standardized beta coefficients.

# 6. Predictions on Unseen Test Data

After fitting the baseline model, the notebook predicts all 40 test observations. It creates a useful comparison table containing TV, Radio, Newspaper, actual Sales, predicted Sales, signed prediction error, and absolute error.

| Test row | Actual | Predicted | Error | Abs. error |
|---|---|---|---|---|
| 1 | 16.9 | 16.41 | +0.49 | 0.49 |
| 2 | 22.4 | 20.89 | +1.51 | 1.51 |
| 3 | 21.4 | 21.55 | -0.15 | 0.15 |
| 4 | 7.3 | 10.61 | -3.31 | 3.31 |
| 5 | 24.7 | 22.11 | +2.59 | 2.59 |
| 6 | 12.6 | 13.11 | -0.51 | 0.51 |
| 7 | 22.3 | 21.06 | +1.24 | 1.24 |
| 8 | 8.4 | 7.46 | +0.94 | 0.94 |
| 9 | 11.5 | 13.61 | -2.11 | 2.11 |
| 10 | 14.9 | 15.16 | -0.26 | 0.26 |

The signed error is **Actual** – **Predicted**. A positive value means the model under-predicted; a negative value means it over-predicted. The table demonstrates why MAE and RMSE are useful aggregate measures: individual errors vary from very small to several thousand units.

# 7. Planned Advertising Budget Prediction

The notebook then uses the trained model to evaluate a concrete business scenario: TV = 150, Radio = 20, Newspaper = 30 (spending units in the dataset's $k convention).

The notebook explicitly shows the arithmetic:

```
Sales = 2.9791 + (0.0447 × 150) + (0.1892 × 20) + (0.0028 × 30) = 2.9791 + 6.7094 + 3.7839 + 0.0828 =
13.5552 thousand units
```

Therefore, the reported prediction is **13.56 thousand units**, approximately **13,555 units**.

**Business interpretation:** this is a model-based scenario estimate, not a guarantee. It is most defensible when the proposed budget lies within the range of historical observations and when the relationship learned from the historical data remains reasonably stable.

# 8. Model Evaluation and What the Metrics Mean

| Metric | Notebook result | Meaning |
|---|---|---|
| MAE | ≈ 1.46 | Average absolute distance between actual and predicted Sales. |
| MSE | ≈ 3.17 | Average squared error; larger errors receive much more weight. |
| RMSE | 1.7816 | Typical error scale in thousand Sales units; easier to interpret than MSE. |
| Test $R^2$ | 0.8994 | About 89.94% of test-set Sales variance is explained. |
| Train $R^2$ | 0.8957 | About 89.57% of training-set variance is explained. |
| Adjusted $R^2$ | ≈ 0.8937 | $R^2$ adjusted for the number of predictors; useful for model comparison. |

The close train/test $R^2$ values are a positive sign: training performance is not dramatically higher than unseen-test performance. In this split, the model does not show an obvious train/test performance gap suggesting severe overfitting.

**Metric hierarchy:** $R^2$ answers “how much variation is explained?”, while MAE/RMSE answer “how large are the prediction errors?”. A strong report should use both types rather than relying on $R^2$ alone.

# 9. Statistical Significance and Standardized Effects

This is one of the strongest additions in the notebook compared with the older PDF. The notebook first standardizes X and y to obtain standardized beta coefficients, then fits an OLS model with statsmodels to obtain p-values, t-statistics and confidence intervals.

| Feature | Raw slope | Std. beta | p-value | Significant at 5%? |
|---|---|---|---|---|
| TV | 0.0447 | 0.7368 | <0.001 | Yes |
| Radio | 0.1892 | 0.5466 | <0.001 | Yes |
| Newspaper | 0.0028 | 0.0110 | 0.6958 | No |

**Key insight:** TV has the largest standardized beta (0.7368), followed by Radio (0.5466). This differs from simply ranking raw coefficients, where Radio's 0.1892 is numerically larger than TV's 0.0447. Standardized coefficients are therefore more appropriate for comparing relative effect strength when predictor scales differ.

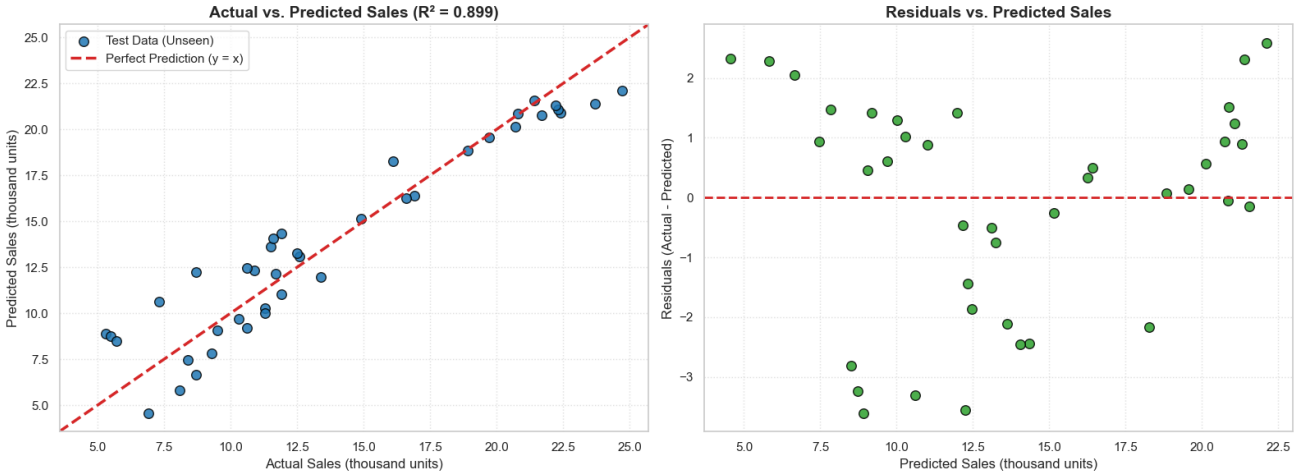
Newspaper's p-value of 0.6958 is far above 0.05 in the notebook's OLS model. That means the notebook does not find sufficient statistical evidence that Newspaper has a non-zero linear effect after TV and Radio are controlled for. This does **not** prove that Newspaper can never affect Sales; it means the current sample and model do not establish a statistically significant partial effect.

The OLS summary reports training $R^2 \approx 0.896$, adjusted $R^2 \approx 0.894$, an F-statistic of 446.6 with an extremely small overall p-value, and a Durbin–Watson statistic of 2.161. The residual normality tests also flag non-normal residual behavior (Omnibus and Jarque–Bera p-values are effectively zero), which is a reason to treat classical inference cautiously and to inspect residuals rather than relying on p-values alone.

# 10. Prediction and Residual Diagnostics

The notebook generates two diagnostic plots side by side. The first compares actual Sales with predicted Sales and overlays the perfect-prediction line y=x. Points close to that line indicate accurate predictions.

The second plot shows residuals (Actual – Predicted) against predicted Sales. A good linear model generally benefits from residuals scattered around zero without an obvious curve or funnel pattern. The plot is therefore a model-assumption check, not merely a presentation graphic.

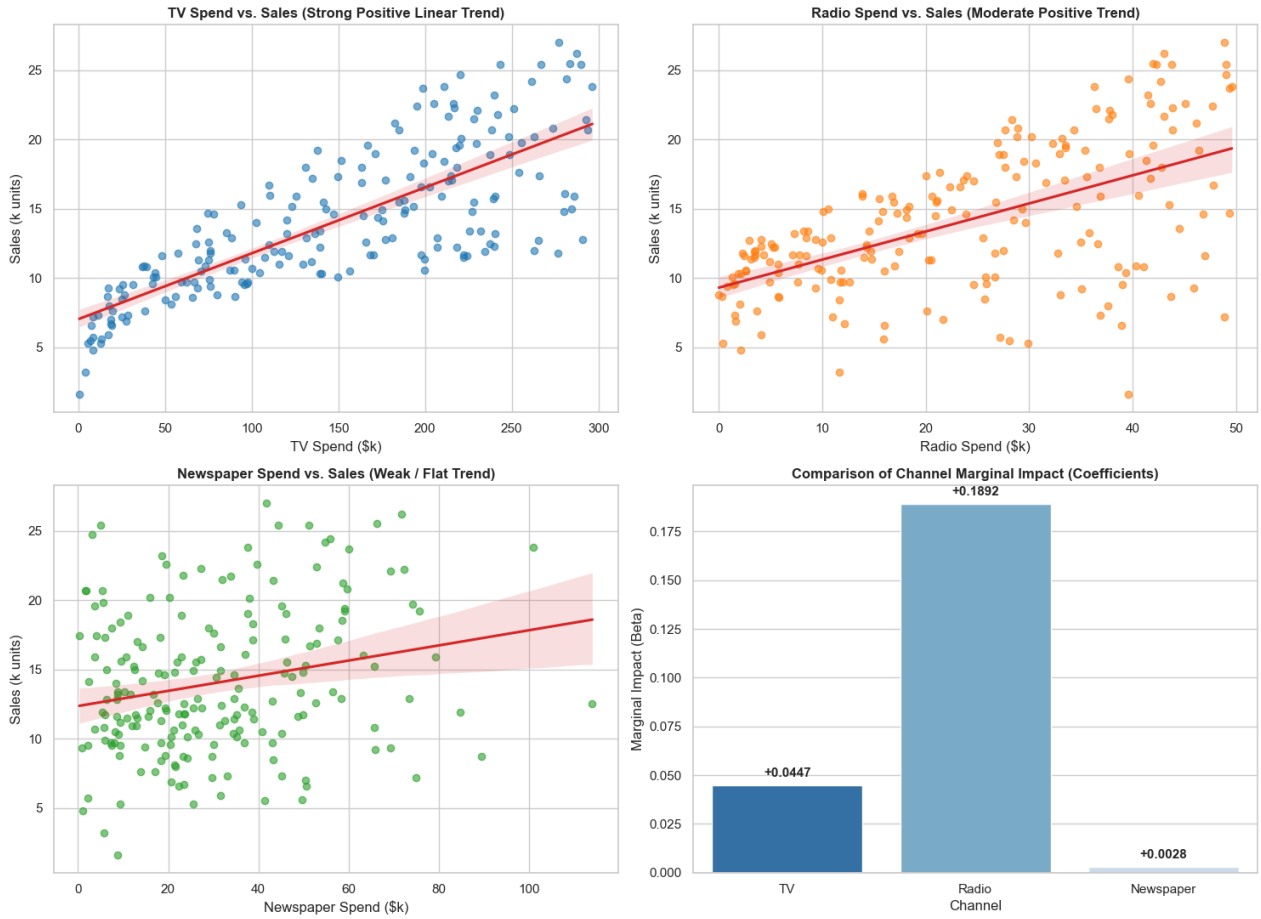



Notebook diagnostic output: actual-vs-predicted and residual plots.

**Better reporting practice:** the residual plot should be checked for curvature, changing spread, and extreme points. The notebook's OLS normality statistics suggest that the residual distribution is not perfectly normal, so the model's predictive usefulness and its inferential assumptions should be discussed separately.

# 11. Feature-to-Sales Relationships

The notebook plots TV vs Sales, Radio vs Sales, and Newspaper vs Sales using regression lines. These plots provide a visual explanation of the correlation results and help identify whether each relationship looks approximately linear.




Notebook output: three feature/Sales regression plots plus a coefficient comparison.

The visual and numerical evidence point in the same general direction: TV has the strongest pairwise association with Sales, Radio also has a meaningful positive relationship, and Newspaper is comparatively weak.

# 12. Empirical Model Improvement: TV × Radio Interaction

The final notebook cell tests a concrete improvement rather than merely recommending one. It creates **TV_Radio_Interaction = TV × Radio** and fits an enhanced regression using TV, Radio and the interaction term.

| Model | Test $R^2$ | RMSE | Change |
|---|---|---|---|
| Baseline Linear Regression | 89.94% | 1.7816 | Reference |
| Enhanced TV×Radio Interaction | 97.43% | 0.9015 | RMSE reduced ≈ 49.4% |

This is a substantial improvement on the same train/test split. The result suggests that the effect of TV advertising may depend on the amount of Radio advertising, rather than the two channels acting as purely independent additive contributions.

**Caution:** the interaction model is evidence of improved predictive performance on this particular split, not proof that it will always outperform the baseline. Cross-validation or repeated train/test splits would provide stronger evidence that the improvement generalizes.

# 13. Consistency Check Against the Existing PDF

The older PDF reports a different fitted equation: **Sales = 4.7141 + 0.0545×TV + 0.1009×Radio + 0.0043×Newspaper**, and a planned-budget prediction of approximately **15.04**. ■ This is not the equation produced by the current notebook execution.

The current notebook instead produces intercept 2.9791, TV 0.0447, Radio 0.1892, Newspaper 0.0028 and planned-budget prediction 13.5552. Therefore, the new report intentionally does **not** reproduce the older PDF's numbers. The notebook is the primary source for this enhanced report.

This difference should be treated as a data/model-version consistency issue. Before submitting a final academic report, it is important to ensure that the notebook, dataset, model outputs and report all come from the same execution.

**Why the difference matters: using the older PDF's coefficients would produce a different business recommendation and a different predicted sales value. A report that mixes figures from two different model runs can look polished but still be technically inconsistent.**

# 14. Recommendations and Next Steps

| Priority | Recommendation | Why it improves the analysis |
|---|---|---|
| 1 | Use the interaction model as a candidate, but validate it with cross-validation. | Confirms whether the 97.43% $R^2$ and 49.4% RMSE reduction generalize. |
| 2 | Investigate Newspaper further before dropping it automatically. | Its p-value is non-significant, but feature usefulness can depend on interactions, nonlinearities and validation performance. |
| 3 | Check residual assumptions systematically. | The OLS normality diagnostics indicate non-normal residual behavior. |
| 4 | Compare against Ridge, Lasso, Random Forest and Gradient Boosting. | Provides a stronger benchmark than comparing only two linear specifications. |
| 5 | Use standardized effects for channel comparison. | TV and Radio have different units/ranges, so raw coefficient size alone can mislead. |
| 6 | Add confidence/prediction intervals for budget scenarios. | A single 13.56k prediction hides uncertainty and gives decision-makers a false sense of precision. |

**Business decision rule:** the model should be used as a decision-support tool. Budget allocation should also consider campaign cost, reach, audience quality, timing, creative quality and diminishing returns—factors that are not represented by this four-column dataset.

# 15. Final Conclusion

The submitted notebook presents a complete baseline regression workflow: it loads and cleans the advertising data, validates the dataset, explores correlations, creates a reproducible train/test split, trains a Multiple Linear Regression model, evaluates predictions, performs statistical inference, visualizes diagnostics, and tests an improved interaction-based specification.

The baseline model explains approximately **89.94%** of test-set Sales variation and has an RMSE of **1.7816 thousand units**. The planned budget scenario produces approximately **13.56 thousand units** of predicted Sales.

The strongest analytical insight is that the final interaction experiment materially improves performance: **97.43% $R^2$** and **0.9015 RMSE**. This makes the notebook more than a simple regression exercise; it demonstrates how model diagnostics and feature engineering can lead to a better predictive specification.

**Submission-ready takeaway:** TV and Radio provide the clearest evidence of useful predictive information, Newspaper is weak and statistically non-significant in the baseline OLS specification, and the TV×Radio interaction is a promising enhancement that should be validated with cross-validation before being used for real budget decisions.

Prepared directly from the executed notebook outputs and checked against the supplied five-page PDF.